Violet Yinuo Han

Ph.D. Student, Human-Computer Interaction Institute, Carnegie Mellon University

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Research Interests

Physical AI • Human-Robot Interaction • User Modeling • Embodied Agents • Distributed Robotics

My research builds physical AI systems that understand and adapt to people in everyday environments. I develop methods that let embodied agents *infer user preferences* and *intent* to *adapt assistance*, and I build *robot embodiments* that bring intelligence into everyday spaces.

Education

Ph.D. Student, Carnegie Mellon University, School of Computer Science — Started August 2023
Human-Computer Interaction Institute, advised by **Prof. Alexandra Ion**

M.S., Carnegie Mellon University — Graduated May 2023
Thesis: Building a Bidirectional Bridge Between the Digital and Physical Worlds

B.F.A., University of Michigan — Graduated May 2020
Thesis: 86 Billions: Experiencing the Human Brain From a Neuron's Perspective

Publications

I have published 9 peer-reviewed publications, 7 as first author, and received 2 best-paper/invited-extension recognitions.

[1] **Priors of U: Dynamic Preference Learning for Embodied Interaction**
Violet Yinuo Han, Alexandra Ion
Proceedings of ACM UIST'26, Detroit, MI. Nov. 2–5, 2026

[2] **Objects with Arms: How Robot Embodiment Shapes Perception in Physical Collaboration**
Violet Yinuo Han, Ziru Wei, Aiden Yiliu Li, Chris Wu, Alexandra Ion
Proceedings of IEEE RO-MAN'26, Kitakyushu, Japan. Aug. 24–28, 2026

[3] **Towards Unobtrusive Physical AI: Augmenting Everyday Objects with Intelligence and Robotic Movement for Proactive Assistance**
Violet Yinuo Han, Jesse T. Gonzalez, Christina Yang, Zhiruo Wang, Scott E. Hudson, Alexandra Ion
Proceedings of ACM UIST'25, Busan, Korea. Sept. 28 – Oct. 1, 2025 (22.2% acceptance rate)

[4] **Transforming Everyday Objects into Dynamic Interfaces using Smart Flat-Foldable Structures**
Violet Yinuo Han, Amber Yinglei Chen, Mason Zadan, Jesse T. Gonzalez, Dinesh K. Patel, Wendy Fangwen Yu, Carmel Majidi, Alexandra Ion
Proceedings of ACM UIST'25, Busan, Korea. Sept. 28 – Oct. 1, 2025 (22.2% acceptance rate)

[5] **A Dynamic Bayesian Network Based Framework for Multimodal Context-Aware Interactions**
Violet Yinuo Han, Tianyi Wang, Hyunsung Cho, Kashyap Todi, Ajoy Savio Fernandes, Andre Levi, Zheng Zhang, Tovi Grossman, Alexandra Ion, Tanya R. Jonker
Proceedings of ACM IUI'25, Cagliari, Italy. March 24–27, 2025 (25% acceptance rate) **Invited Journal Extension**

[6] **Robotic Metamaterials: A Modular System for Hands-On Configuration of Ad-Hoc Dynamic Applications**
Zhitong Cui, Shuhong Wang, **Violet Yinuo Han**, Tucker Rae-Grant, Willa Yunqi Yang, Alan Zhu, Scott E. Hudson, Alexandra Ion
Proceedings of ACM CHI'24, Honolulu, HI. May 11–16, 2024 (26.4% acceptance rate)

[7] **Parametric Haptics: Versatile Geometry-based Tactile Feedback Devices**
Violet Yinuo Han, Abena Boadi-Agyemang, Yuyu Lin, David Lindlbauer, Alexandra Ion
Proceedings of ACM UIST'23, San Francisco, CA. Oct. 29 – Nov. 1, 2023 (21% acceptance rate)

[8] **BlendMR: A Computational Method to Create Ambient Mixed Reality Interfaces**
Violet Yinuo Han, Hyunsung Cho, Kiyosu Maeda, Alexandra Ion, David Lindlbauer
Proceedings of ACM ISS'23, Pittsburgh, PA. Nov. 5–8, 2023 (24–28% acceptance rate) **Best Paper Award, Top 1**

[9] **Permeable Thermistor Temperature Sensors Based on Porous Melamine Foam**
Hugo de Souza Oliveira, Niloofar Saeedzadeh Khaanghah, **Violet Yinuo Han**, Alejandro Carrasco-Pena, Alexandra Ion, Michael Haller, Giuseppe Cantarella, Niko Münzenrieder
IEEE Sensors Letters, Vol. 7, No. 5, pp. 1–4, Art. no. 2500904, May 2023

Awards and Grants

Qualcomm Innovation Fellowship, $100k, with Weiwei Sun — 2026
Special Recognition for Outstanding Review, ACM UIST — 2025, 2026
Best Paper Award, Top 1, ACM ISS'23 for *BlendMR* [8] — 2023
Snap Creative Challenge Award, $10k, with Prof. David Lindlbauer and Dr. Yukang Yan — 2022
Carnegie Mellon University SoA Commendation — 2022
Carnegie Mellon University SoA Merit Scholarship, $10k — 2021
University of Michigan University Honors — 2018, 2020

Professional Experience

Research Scientist Intern, Meta Reality Labs — May – Sept. 2024
Proposed a dynamic Bayesian network framework for multimodal context-aware interaction. Published at IUI'25 [5] and invited for journal extension.

System Experience Design Intern, OnePlus — May – July 2021
Worked with designers and engineers to maintain and update OnePlus OxygenOS.

Interaction Design Intern, WhaleDynamic — Feb. – May 2021
Designed interactions for an autonomous delivery vehicle and presented a prototype at the Shanghai International Automobile Industry Exhibition.

Invited Talks

CMU Robotic Caregiving and Human Interaction Lab, hosted by Prof. Zackory Erickson — April 2026
The Ellis School, hosted by Prof. Tai Sing Lee — April 2025

Research Demos

CMU Robotics Innovation Center Opening, Towards Unobtrusive Physical AI [3] — Pittsburgh, 2026
CMU XRTC Symposium, Towards Unobtrusive Physical AI [3] — Pittsburgh, 2025
ACM UIST, Towards Unobtrusive Physical AI [3] — Busan, 2025
CMU XRTC Symposium, Parametric Haptics [7] — Pittsburgh, 2024
ACM UIST, Parametric Haptics [7] — San Francisco, 2023

Media Coverage

ComputerWorld, "Why ambient robots beat humanoid robots" [3] — 2026
CNET, "Robot Coffee Cups? Self-Driving Trivets? AI Researchers Made It Happen" [3] — 2025
CMU News, "CMU Researchers Use AI To Turn Everyday Objects Into Proactive Assistants" [3] — 2025
TechXplore, "A stapler that knows when you need it: Using AI to turn everyday objects into proactive assistants" [3] — 2025

The Brighter Side of News, “AI revolution: Researchers teach everyday objects to sense, think, and move” [3] 2025

Teaching Experience

Teaching Assistant, Carnegie Mellon University Aug. – Dec. 2026

Human-AI Interaction, Prof. Sherry Wu and Prof. Sauvik Das

Guest Lecturer, Carnegie Mellon University April 2025

Computational Methods for Interactive Systems (05-499/899), Prof. Alexandra Ion. Lecture on using dynamic Bayesian networks to model interactions for building interactive systems.

Teaching Assistant, Carnegie Mellon University Aug. – Dec. 2022

Computational Perception (15-387/86-375/675), Prof. Tai Sing Lee. Hosted weekly office hours, created and graded assignments, and wrote lecture notes to support student learning.

Mentoring

Danny Linitz, Design at CMU, Undergraduate Research Assistant Fall 2026
Selina Lee, Design at CMU, Undergraduate Research Assistant Fall 2026
Dian Zhu, Mathematics at CMU, Undergraduate Research Assistant Fall 2025
Narayan Ashanahalli, Architecture at CMU, Masters Research Assistant Fall 2025
Aiden Li, Computer Science at UCL, Undergraduate Research Assistant Summer 2025
Christina Yang, Design and HCI at CMU, Masters Research Assistant Fall 2024
Anagha Srinivasan, Computer Science at CMU, Undergraduate Research Assistant Spring 2024
Ziyu Li, ECE at CMU, Undergraduate Research Assistant Fall 2023
Wendy Yu, Design and HCI at CMU, Undergraduate Research Assistant Fall 2023, Spring 2024

Academic Service

Poster Area Chair, ACM UIST 2026
Reviewer, PAMHCI (ISS) 2026
Reviewer, ACM UIST 2024, 2025, 2026
Reviewer, ACM CHI 2024, 2025
Reviewer, ACM CHI Late-Breaking Work 2023, 2024
Assistant to Session Chair, ACM CHI 2024
Student Volunteer, ACM UIST 2022